

Anshul Kumar Yadav

☎ +91 7357499994 @ anshul.4ky@gmail.com 📍 Mumbai, India
in LinkedIn 🎓 Google Scholar

PROFESSIONAL SUMMARY

Currently, I work as **Project Research Assistant** at **Koita Centre for Digital Health, IIT-Bombay**, concentrating on developing artificial intelligence frameworks for healthcare use cases. I have research experience in computational intelligence, computer vision, and optimisation frameworks.

EDUCATION

Indian Institute of Technology Bombay

Masters by Research in Healthcare Informatics; GPA: 9.09/10.00 (Distinction)

Research Focus: **Artificial intelligence frameworks for healthcare use cases.**

Mumbai, India

August 2026 – Pursuing

Swami Keshvanand Institute of Technology Management & Gramothan

B.Tech. in Electrical Engineering; GPA: 9.09/10.00 (Distinction)

Research Focus: **Smart Grid Technology and Applications**

Jaipur, India

August 2019 – July 2023

SKILLS

Technical: Python (Plotly, PyTorch, HuggingFace), Git, Docker, MATLAB, Latex, Edge Impulse, MS Office (MS Excel, MS Outlook, MS PowerPoint, MS OneNote), Notion, GPU platform (DGX Server), Linux

Bibliographical software: VosViewer, SciMAT, Bibliometrix

Deployment boards: Esp32, Nano 33 BLE, Raspberry Pi, Jetson Nano

Soft: Ability to work in diverse environments, handle multi-disciplinary projects, problem-solver, pro-active, willing to learn outside the comfort zone, good leadership, communication, mentoring, and presentation skills.

RESEARCH EXPERIENCE

Koita Centre for Digital Health (KCDH)

Project Research Assistant

Bombay, India

July 2025 – Present

- **MammoQ: Mammography Assessment Pipeline**

- * Developed a modular mammography engine that includes DICOM ingestion, preprocessing, abnormality detection, FGT segmentation, and ACR breast-density grading.
- * Fine-tuned vision models for breast, pectoral, artifact, calcification, lymph-node, and mass detection across mammography views.
- * Fine-tuned vision-language models for breast-density assessment and ACR classification, combining segmentation outputs with clinically guided prompting.
- * Implemented rule-based clinical refinement modules for mass-fusion, artefact-overlap filtering, anatomical lymph-node relabelling, calcification association, and side-level Normal/Abnormal decisions.
- * Built production-oriented orchestration with centralised YAML configuration, resumable patient manifests, structured logging, Prometheus metrics, and Observability dashboards.

Computer Science & Engineering, IITB

Project Technical Assistant

Bombay, India

April 2025 – June 2025

- **Image Quality Assessment Tool**

- * Implemented lightweight classification models using TinyViT and MobileViT architectures, ensuring efficient inference suitable for deployment in low-resource healthcare settings.
- * Explored semi-supervised learning techniques to reduce the annotation burden.

CSIR-Central Electronics Engineering Research Institute, Govt. of India

Junior Research Fellow

Pilani, India

August 2023 – March 2025

- **Revitalizing Ancient Murals Using Deep-Learning Techniques**

- * Performed data acquisition at Mandawa Kothi and image preprocessing to study realistic mural damage.
- * Developed a damage segmentation framework employing GANs and an ensemble network to identify varying levels of damage in Shekhawati murals.

- * Studied and implemented inpainting models including CNN-, GAN-, and Transformer-based networks to identify research gaps.
- * Developed an algorithm integrating a damage segmentation model, visual feature-based clustering, and modular inpainting for refined results.
- **Battery Management System for Redox Flow Battery (RFB)**
 - * Studied the operation and energy management of an industrial-grade 5kWh RFB battery.
 - * Developed a regression model for state-of-charge prediction, including feature selection and dataset augmentation.
 - * Deployed TinyML models to Arduino IoT boards using TensorFlow Lite.
 - * Integrated node-edge-cloud computing paradigms for enhanced SoC estimation.
- **Additive Manufacturing Anomaly Detection**
 - * Designed an image- and sensor-based anomaly detection framework with hardware integration.
 - * Collected vibration data using MPU 6050 sensors on two distinct 3D printing setups.
 - * Processed vibration signals to distinguish failure types and developed a computationally efficient classifier.
- **X-ray AI: Baggage Security System**
 - * Built an AI-enhanced baggage scanner using KitiScan 6040 with YOLOv11 for real-time threat detection.
 - * Identified prohibited items like weapons, tools, and sharp objects in X-ray imagery.
 - * Enabled threat object recognition in video mode, extending functionality for recorded footage.

Raman Lab

Research Intern

Jaipur, India

June 2022 – August 2023

- Developed a hybrid optimization algorithm for a multi-objective smart home energy management system to minimize electricity costs without compromising comfort.
- Designed and simulated a renewable-integrated microgrid with PV, wind, battery, genset, and V2G systems. Used metaheuristics to minimize cost and avoid load curtailment.
- Optimized parameters for a semi-empirical battery degradation model to estimate Li-ion battery state-of-health.
- Created a networked microgrid using Australian grid data and implemented fog-cloud architecture for data processing and uncertainty analysis.
- Applied Q-learning with a novel two-stage reward function to optimize household energy usage, ensuring convergence and constraint adherence.

PUBLICATIONS

Journal Publications

7. **A. K. yadav**, et al. “A Systematic Review on Advancing Agricultural Applications with Machine Learning assisted Cold Atmospheric Plasma.” **Revision**, Computers and Electronics in Agriculture.
6. **A. K. yadav**, et al. “Reconstructing degraded areas of old Indian Wall Paintings through Image Inpainting.” **Accepted**, International Journal of Arts and Technology.
5. **A. K. yadav**, et al. “Assessment of Deep Learning algorithms for Damage Segmentation in Indian Murals.” **Accepted**, International Journal of Arts and Technology.
4. A. K. Saini, **A. K. yadav**, et al. “**A Comprehensive review on technological breakthroughs in precision agriculture: IoT and emerging data analytics.**”, European Journal of Agronomy.
3. **A. K. yadav**, et al. “A Systematic Review on Energy Management for Redox Flow Batteries via Intelligent Data Processing.” **Accepted**, Energy Storage.
2. **A. K. yadav**, et al. “Deep learning-based framework for damage introduction and segmentation in ancient wall paintings.” **Accepted**, Sensing and Imaging.
1. S. Yelisetti, **A. K. yadav**, et al. “Application of Sine Cosine Weighted Mean of Vector Optimization Algorithm for Optimal Energy Consumption Cost in Residential Buildings.” **In Review**, Journal of Building Engineering.

Conference Publications

20. A. Ranjan, ..., **A. K. yadav**, et al. “Evaluation of a ± 5 kV Solid-state-based 200 ns duration Pulse Generator.” **Accepted**, *1st International Conference on Power Electronics Converters in Transportation and Energy Application*

19. V. S. Mahala, **A. K. yadav**, et al. "Emission Based Energy Management Framework for Networked Microgrids using Edge-Fog Computing." **Accepted**, 2025 5th IEEE International Conference on Sustainable Energy and Future Electric Transportation (SEFET)
18. **A. K. yadav**, et al. "Generalisable Predictive Maintenance in Additive Manufacturing using Machine Learning." **Accepted**, 4th IEEE International Conference on Innovative Sustainable Computational Technologies
17. **A. K. yadav**, et al. "Detecting Additive Manufacturing Anomalies with a Shallow Residual Convolution Network." **Accepted**, 2024 Unified International Conference on Emerging Technologies in Cyber-Physical Systems and Industrial AI (Unified-2024)
16. **A. K. yadav**, et al. "Computing Framework for SoC Estimation using On-device Learning." **Accepted**, 2024 IEEE 11th Power India International Conference (PIICON)
15. **A. K. yadav**, et al. "Edge Computing enabled Battery State of Charge estimation using Tiny Machine Learning techniques." **Accepted**, 2024 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)
14. V. S. Mahala, **A. K. yadav**, et al. "Bi-Level Optimization Framework for Energy Management in Networked Microgrid Using Edge-Fog Computing Paradigm." **Accepted**, 2024 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)
13. **A. K. yadav**, et al. "Machine learning based data-driven approach for state-of-charge estimation in redox flow battery." 2024 15th International IEEE Conference on Computing Communication and Networking Technologies (ICCCNT)
12. A. M. Dharwal, **A. K. yadav**, et al. "Steel surface defect detection using machine learning techniques." 2024 IEEE International Conference on Electronics, Communication and Signal Processing (ICESP)
11. V. S. Mahala, **A. K. yadav**, et al. "Uncertainty Estimation of PV and Load Using Deep Learning for Networked Microgrid." 2024 4th IEEE International Conference on Sustainable Energy and Future Electric Transportation (SEFET)
10. **A. K. yadav**, et al. "Uncertainty aware State-of-Charge estimation for Li-ion Batteries using Deep Learning." 2023 2nd IEEE International Conference on Measurement, Instrumentation, Control, and Automation (ICMICA)
9. **A. K. yadav**, et al. "An insight into energy management through smart data analytics for redox flow battery." **Conference abstract**, 2024 International Conference on Sustainable Energy and Environment (IC-SEE)
8. **A. K. yadav**, et al. "Revitalizing ancient murals in the Shekhawati region through image inpainting techniques." 2024 11th IEEE International Conference on Signal Processing and Integrated Networks (SPIN)
7. A. Saxena, **A. K. yadav**, et al. "Seven Sisters Optimization Algorithm." 2023 2nd IEEE International Conference on Futuristic Technologies (INCOFT)
6. K. Baberwal, **A. K. yadav**, et al. "Data Driven Energy Management of Residential PV-Battery System Using Q-Learning." 2023 IEEE International Conference on Recent Advances in Systems Science and Engineering (RASSE)
5. V. S. Mahala, **A. K. yadav**, et al. "Networked Hybrid AC-DC Microgrids: Leveraging Fog Computing and Linear Solver for Efficient Energy Management." 2023 IEEE International Conference on Recent Advances in Systems Science and Engineering (RASSE)
4. N. Kumar, **A. K. yadav**, et al. "Battery Energy Storage Sizing and Operational Strategy for Microgrid Considering Electric Vehicle." 2023 3rd IEEE International Conference on Sustainable Energy and Future Electric Transportation (SEFET)
3. S. Bharti, ..., **A. K. yadav**, et al. "Optimal Parameter Estimation of CAPN Model for Li-ion Battery." 2023 IEEE International Conference on Computer, Electronics & Electrical Engineering & their Applications (IC2E3)
2. **A. K. yadav**, A. Sharma, et al. "Optimization Scheme for Power Transmission in Wireless Sensor Network." 2023 IEEE International Conference on Power, Instrumentation, Energy and Control (PIECON)
1. V. K. Saini, **A. K. yadav**, et al. "Multi-Agent based Cloud Energy Storage Framework for Residential Community." 2022 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)

AWARDS, ACHIEVEMENTS, & CERTIFICATES

Examination: UGC-National Eligibility Test (Dec-2024, 85.03%)

Best Paper Awards: at PIECON-(2023) for undergraduate project, ICMICA-(2024)

Best Poster Awards: SEFET-(2024)

IELTS Score: 7.5/9 (Listening: 7.5, Reading: 8.5, Writing: 7.0, Speaking: 7.0)

LEAD1x: Exercising Leadership: Foundational Principles: Funded by Aspire Institute, Harvard University

Positions: Student editor at Skit Times (College Magazine, 2019-2022), IEI student member (2019-2023)

LinkedIn badges: Machine Learning, MATLAB, GitHub, Docker

Certifications: Python, MATLAB, Simulink, Metaheuristic Optimization, Data Science